



**KEMENTERIAN PENDIDIKAN TINGGI, SAINS,
DAN TEKNOLOGI
POLITEKNIK NEGERI MALANG
JURUSAN TEKNOLOGI INFORMASI**

Jl. Soekarno Hatta No.9 Jatimulo, Lowokwaru, Malang, 65141
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**FINAL EXAM OF EVEN SEMESTER OF THE 2025/2026 ACADEMIC
YEAR**

Course : Practicum Algorithms and Data Structures
Day/ date : Friday, June 6, 2026
Time : 08.00 – 18.00
Class :

Provision general :

- 1) n't any provision using array, linked list or double linked list. Will be given addition points based on level difficulty . Array does not There is additional , linked list +10, double linked list +15.
- 2) Not allowed using the Java Collection Framework.

Question

For NIM ending in number odd

Make it a program for do transaction purchase items used by the cashier . Some provisions that must be met is in the program is as following

1. Program created use menu system and special item data Already content when the program is run . (10)
2. Minimum consists of from 2 classes , namely Items (code , name, price , stock) and Transactions (code, quantity) goods, total price). In the Transaction class must related with the class of Goods. Special code transaction will automatic increment sequence . (5)
3. the transaction data input process is entered is only code goods and quantity . Purchasing process result in stock decreases at every goods and displays the nominal amount that must be paid . Then after Once the transaction input is complete , then display code transaction , item name , price unit , quantity , and total price . Then after a number of The criteria that are also met are ; (40)
 - a. When stock remaining 0 means No can done purchase goods the
 - b. When the total payment > 500000 will get 5% discount
 - c. If the member will get piece price 2%
4. The program can display all over previous transactions and displays the total income . (25)
5. The program can sort all over previous transactions based on the name of the item ascending . (20)



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Question

For NIM ending in number even

Create a program for serve queue payment tax vehicle motorized . Some provisions that must be met filled is as following

1. Program created use menu system and special vehicle data Already content when the program is run . (10)
2. Minimum consists of from 2 classes , namely Vehicle (TNKB, name, type , cc, year , month must pay) and Tax (code , nominal, fine , month pay). In the Tax class, you must related with Vehicle class . Special the code in the Tax class will automatic increment sequence . (5)
3. tax data is entered is only TNBK and the moon pay . Some provision related with type , cc, and year is as following : (40)
 - a. Types consist of 2 wheels and 4 wheels
 - b. For 2 cc wheels , it consists of less than 100, between 100 to 250, and more than 250. Tax rates in a way The order of cc is 100000, 250000, and 500000.
 - c. For 4 cc wheels , it consists of less than 1000, between 1000 to 2500, and more than 2500. Tax rates in a way The order of cc is 750000, 1000000, and 1500000.
 - d. Charged fine when month pay > month must pay . when 1-3 months late charged fine of 50,000, while > 3 months will multiplied with 50000.
4. The program can display all over previous transactions and displays the total income . (25)
5. The program can sort all over previous transactions based on the name of the item ascending . (20)